



1 Bernoulli Distribution

Property	Formula / Value
PMF	$P(X = 1) = p, \quad P(X = 0) = 1 - p$
Expected Value	p
Variance	$p(1 - p)$
MGF	$1 - p + pe^t \rightarrow q + pe^t$
First Moment	p
Second Moment	p
Cumulants	$\kappa_1 = p, \quad \kappa_2 = p(1 - p)$
Example	Single coin toss

Table 1: Characteristics of the Bernoulli Distribution

- Trials: Single
- Outcomes: Success (1) or Failure (0)
- Parameter: p = Probability of success
- Mean: p
- Variance: $p(1 - p)$

Examples

1. If $p = 0.7$, find $E(X)$ and $Var(X)$.
2. If $p = 0.4$, find $P(X = 1)$.
3. Use MGF to show that $E(X) = p = E(X^2)$



2 Binomial Distribution

The probability distribution is give by the following formula:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

where:

- n = number of trials
- x = number of successes
- p = probability of success

Property	Formula / Value
PMF	$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, x = 0, 1, \dots, n$
Mean	np
Variance	$np(1 - p)$
MGF	$(1 - p + pe^t)^n$
First Moment	np
Second Moment	$np(1 + (n - 1)p)$
Cumulants	$\kappa_1 = np, \quad \kappa_2 = np(1 - p)$
Example	Number of heads in n coin tosses

Table 2: Characteristics of the Binomial Distribution

Examples

- (i) Find $P(X = 3)$ for $n = 5, p = 0.6$.

Solution:

$$P(X = 3) = \binom{5}{3} (0.6)^3 (0.4)^2 = 0.3456$$

- (ii) Find $E(X)$ and $Var(X)$ for $n = 10, p = 0.2$.

Solution:

$$E(X) = 2, Var(X) = 1.6$$

- (iii) ACME Inc. produces super-tactile widgets for the latest DIY craze, and they are sold in boxes of 8. Unfortunately, 2% of them are faulty. Find the probability that a given box has at least one faulty widget in it.

Solution:

Let the probability of a widget being faulty be $p = 0.02$. Since each box contains 8 widgets, we define X as the number of faulty widgets in a box. We assume that each widget is faulty independently with probability p .

The probability that a given widget is not faulty is:

$$P(\text{not faulty}) = 1 - p = 1 - 0.02 = 0.98.$$

Since there are 8 widgets in a box, the probability that all of them are not faulty is:

$$P(\text{no faulty widgets}) = (0.98)^8.$$

Thus, the probability that at least one widget is faulty is:

$$\begin{aligned} P(\text{at least one faulty widget}) &= 1 - P(\text{no faulty widgets}) \\ &= 1 - (0.98)^8. \end{aligned}$$

Computing the numerical value:

$$(0.98)^8 \approx 0.8509, \\ P(\text{at least one faulty widget}) \approx 1 - 0.8509 = 0.1491.$$

Thus, the probability that a given box has at least one faulty widget is approximately 0.1491 or 14.91%.

- (iv) A fair coin is flipped 10 times. What is the probability of getting exactly 6 heads?

Solution:

$$P(X = 6) = \binom{10}{6} (0.5)^6 (0.5)^4 \\ = \frac{10!}{6!(4!)} (0.5)^{10} \\ = 210 \times (0.000976) = 0.205$$

- (v) A factory produces light bulbs, and 90% of them are not defective. If a random sample of 15 bulbs is selected, what is the probability that exactly 12 of them are not defective?

Solution:

$$P(X = 12) = \binom{15}{12} (0.9)^{12} (0.1)^3 \\ = \frac{15!}{12!(3!)} (0.9)^{12} (0.1)^3 \\ = 455 \times (0.282) \times (0.001) = 0.128$$

- (vi) A factory produces light bulbs with a 5% defect rate. If 20 bulbs are tested, what is the probability that exactly 2 are defective?

Solution: This is a **Binomial distribution** problem because:

- Fixed number of trials ($n = 20$),
- Independent trials,
- Probability of success ($p = 0.05$) is constant.

The PMF is:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Substitute $n = 20$, $p = 0.05$, and $k = 2$:

$$P(X = 2) = \binom{20}{2} (0.05)^2 (0.95)^{18} = 190 \cdot 0.0025 \cdot 0.3972 = 0.1887$$

- (vii) A multiple-choice test has 10 questions, each with 4 choices. If a student guesses randomly, what is the probability of getting exactly 3 correct answers?

Solution: This is a **Binomial distribution** problem because:

- Fixed number of trials ($n = 10$),
- Independent trials,
- Probability of success ($p = 0.25$) is constant.

The PMF is:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Substitute $n = 10$, $p = 0.25$, and $k = 3$:

$$P(X = 3) = \binom{10}{3} (0.25)^3 (0.75)^7 = 120 \cdot 0.015625 \cdot 0.1335 = 0.2503$$

- (viii) There were ten green bottles sitting on the wall. The probability of a green bottle accidentally falling is 0.95. What is the probability that fewer than 8 of the green bottles accidentally fall? **8. Fewer than 8 bottles fall**

$$X \sim B(10, 0.95)$$

Find $P(X < 8) = P(X \leq 7)$:

$$P(X \leq 7) = 1 - P(X = 8) - P(X = 9) - P(X = 10)$$

$$P(X = 8) \approx 0.074, \quad P(X = 9) \approx 0.265, \quad P(X = 10) \approx 0.599$$

$$P(X \leq 7) = 1 - (0.074 + 0.265 + 0.599) = 0.062$$

- (ix) Thirty percent of students at a large university are known to be short-sighted. If twenty students are picked at random, find the probability that at most two of them are short-sighted.

Given:

$$X \sim B(20, 0.3)$$

Find $P(X \leq 2)$:

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

Using the binomial formula:

$$P(X = k) = \binom{20}{k} (0.3)^k (0.7)^{20-k}$$

$$P(X = 0) \approx 0.0008, \quad P(X = 1) \approx 0.0074, \quad P(X = 2) \approx 0.0357$$

$$P(X \leq 2) \approx 0.0008 + 0.0074 + 0.0357 = 0.0439$$

- (x) An unbiased coin is tossed 10 times. Find the probability that there are fewer than 3 heads tossed.

$$X \sim B(10, 0.5)$$

Find $P(X < 3) = P(X \leq 2)$:

$$P(X = 0) \approx 0.00098, \quad P(X = 1) \approx 0.0098, \quad P(X = 2) \approx 0.0439$$

$$P(X \leq 2) = 0.00098 + 0.0098 + 0.0439 = 0.0547$$

- (xi) A farmer plants 12 saplings. On average, 15% of saplings planted fail to survive their first winter. Find the probability that more than one of his saplings will die in that first winter. **3. More than one sapling dies**

$$X \sim B(12, 0.15)$$

Find $P(X > 1) = 1 - P(X \leq 1)$:

$$P(X = 0) \approx 0.138, \quad P(X = 1) \approx 0.292$$

$$P(X > 1) = 1 - (0.138 + 0.292) = 0.57$$

- (xii) ACME Inc. produces super-tactile widgets for the latest DIY craze, and they are sold in boxes of 8. Unfortunately, 2% of them are faulty. Find the probability that a given box has at least one faulty widget in it.

$$X \sim B(8, 0.02)$$

$$P(X \geq 1) = 1 - P(X = 0)$$

$$P(X = 0) \approx 0.850$$

$$P(X \geq 1) = 1 - 0.850 = 0.15$$

- (xiii) At Covington High School, 20% of sixth-form mathematics students retake their C1 module. There are 15 students in Mr. Smither's class; find the probability that precisely four of them retake this module.

$$X \sim B(15, 0.2)$$

Find $P(X = 4)$:

$$P(X = 4) = \binom{15}{4} (0.2)^4 (0.8)^{11}$$

$$P(X = 4) \approx 0.218$$

- (xiv) Brian enjoys his pizza. So much so that he eats pizza every night. He likes both pepperoni and mushroom, and each night he chooses randomly and independently between these two options. Given that each night he picks the mushroom option with probability 0.35, find the probability that he eats between 2 and 4 (inclusive) mushroom pizzas this week (7 days).

$$X \sim B(7, 0.35)$$

Find $P(2 \leq X \leq 4)$:

$$P(X = 2) \approx 0.290, \quad P(X = 3) \approx 0.271, \quad P(X = 4) \approx 0.137$$

$$P(2 \leq X \leq 4) = 0.290 + 0.271 + 0.137 = 0.698$$



3 Negative Binomial Distribution

The negative binomial probability formula is:

$$P(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$$

Property	Formula / Value
PMF	$P(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$
Support	$X \in \{r, r+1, r+2, \dots\}$
Mean	$\frac{r}{p}$
Variance	$\frac{r(1-p)}{p^2}$
MGF	$\left(\frac{pe^t}{1-(1-p)e^t} \right)^r$
First Moment	$\frac{r}{p}$
Cumulants	$\kappa_1 = \frac{r}{p}, \quad \kappa_2 = \frac{r(1-p)}{p^2}$
Example	Number of trials needed to achieve r successes

Table 3: Characteristics of the Negative Binomial Distribution

- Counts trials until r -th success
- Success probability: p
- Mean: $\frac{r}{p}$
- Variance: $\frac{r(1-p)}{p^2}$

Examples

- (i) Find $P(X = 5)$ for $r = 2, p = 0.3$.

Solution: $P(X = 5) = \binom{4}{1}(0.3)^2(0.7)^3 = 0.1323$

- (ii) Find $E(X)$ and $Var(X)$ for $r = 3, p = 0.5$.

Solution: $E(X) = 6, Var(X) = 6$

- (iii) A sales representative closes a deal with a probability of 0.3 on each call. What is the probability that he closes his fourth deal on his tenth call?

- (iv) A basketball player makes 60% of their free throws. What is the probability that they make their third successful free throw on the sixth attempt?

Solution:

$$\begin{aligned} P(X = 6) &= \binom{5}{2}(0.6)^3(0.4)^3 \\ &= \frac{5!}{2!(3!)}(0.216)(0.064) = 0.138 \end{aligned}$$

- (v) A researcher is conducting an experiment where the probability of success in each trial is 0.3. What is the probability that the 5th success occurs on the 10th trial?

Solution:

$$\begin{aligned} P(X = 10) &= \binom{9}{4}(0.3)^5(0.7)^5 \\ &= 126 \times (0.000408) = 0.0514 \end{aligned}$$

- (vi) A researcher is conducting experiments until they get 3 successful results. Each experiment has a 20% chance of success. What is the probability that the 3rd success occurs on the 7th trial?

Solution: This is a **Negative Binomial distribution** problem because:

- We are counting the number of trials ($k = 7$) needed to achieve a fixed number of successes ($r = 3$),
- Probability of success ($p = 0.2$) is constant.

The PMF is:

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

Substitute $r = 3, p = 0.2$, and $k = 7$:

$$P(X = 7) = \binom{6}{2}(0.2)^3(0.8)^4 = 15 \cdot 0.008 \cdot 0.4096 = 0.0492$$

- (vii) A basketball player has a 70% chance of making a free throw. What is the probability that they will make their 5th successful free throw on the 8th attempt?

Solution: This is a **Negative Binomial distribution** problem because:

- We are counting the number of trials ($k = 8$) needed to achieve a fixed number of successes ($r = 5$),
- Probability of success ($p = 0.7$) is constant.

The PMF is:

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

Substitute $r = 5, p = 0.7$, and $k = 8$:

$$P(X = 8) = \binom{7}{4}(0.7)^5(0.3)^3 = 35 \cdot 0.16807 \cdot 0.027 = 0.1597$$

- (viii) A sales representative closes a deal with a probability of 0.3 on each call. What is the probability that he closes his **fourth** deal on his **tenth** call?

Solution: Using the Negative Binomial PMF:

$$\begin{aligned}
 P(X = 10) &= \binom{9}{3} (0.3)^4 (0.7)^6 \\
 &= \frac{9!}{3!(6!)} \times (0.3)^4 \times (0.7)^6 \\
 &= 84 \times 0.0081 \times 0.1176 \\
 &= 0.080
 \end{aligned}$$

Thus, the probability is ****0.080****.

- (ix) A basketball player has a 60% chance of making a free throw. What is the probability that she makes her **fifth** successful shot on her **eighth** attempt?

Solution: The probability mass function (PMF) of a Negative Binomial distribution is:

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

where: - $r = 5$ (number of successes), - $k = 8$ (total attempts), - $p = 0.6$ (success probability).

$$\begin{aligned}
 P(X = 8) &= \binom{7}{4} (0.6)^5 (0.4)^3 \\
 &= \frac{7!}{4!(3!)} \times (0.6)^5 \times (0.4)^3 \\
 &= 35 \times 0.07776 \times 0.064 \\
 &= 0.174
 \end{aligned}$$

Thus, the probability is ****0.174****.



4 Geometric Distribution

The geometric probability formula is:

$$P(X = x) = (1-p)^{x-1} p$$

Property	Formula / Value
PMF	$P(X = x) = p(1-p)^{x-1}, x = 1, 2, 3, \dots$
Expected Value	$\frac{1}{p}$
Variance	$\frac{1-p}{p^2}$
MGF	$\frac{pe^t}{1-(1-p)e^t}$
First Moment	$\frac{1}{p}$
Cumulants	$\kappa_1 = \frac{1}{p}, \quad \kappa_2 = \frac{1-p}{p^2}$
Example	Number of trials until first success

Table 4: Characteristics of the Geometric Distribution

- Counts trials until first success
- Success probability: p
- Mean: $\frac{1}{p}$
- Variance: $\frac{1-p}{p^2}$

Examples

- (i) Find $P(X = 4)$ when $p = 0.2$.

Solution: $P(X = 4) = (0.8)^3(0.2) = 0.1024$

- (ii) Find $E(X)$ and $Var(X)$ when $p = 0.4$.

Solution: $E(X) = 2.5$, $Var(X) = 3.75$

- (iii) A die is rolled repeatedly until a six appears. What is the probability that the first six appears on the fourth roll?

Solution:

$$\begin{aligned} P(X = 4) &= \left(1 - \frac{1}{6}\right)^3 \times \frac{1}{6} \\ &= (5/6)^3 \times (1/6) = 0.096 \end{aligned}$$

- (iv) A call center receives calls randomly, and the probability that a call is from a new customer is 0.2. What is the probability that the first call from a new customer occurs on the fifth call?

Solution:

$$P(X = 5) = (0.8)^4(0.2) = 0.0819$$

- (v) A factory produces defective bulbs with a probability of 0.05. What is the probability that the first defective bulb is found on the 7th inspection?

Solution:

$$P(X = 7) = (0.95)^6(0.05)$$

$$= 0.7351 \times 0.05$$

$$= 0.0368$$

Thus, the probability is ****0.0368****.

- (vi) A lightbulb has a 10% chance of burning out each day. What is the probability that it will burn out on the 5th day?

Solution: This is a **Geometric distribution** problem because:

- We are counting the number of trials ($k = 5$) until the first success (burning out),
- Probability of success ($p = 0.1$) is constant.

The PMF is:

$$P(X = k) = (1 - p)^{k-1}p$$

Substitute $p = 0.1$ and $k = 5$:

$$P(X = 5) = (0.9)^4(0.1) = 0.6561 \cdot 0.1 = 0.0656$$

- (vii) A fisherman has a 15% chance of catching a fish each time he casts his line. What is the probability that he will catch his first fish on the 4th cast?

Solution: This is a **Geometric distribution** problem because:

- We are counting the number of trials ($k = 4$) until the first success (catching a fish),
- Probability of success ($p = 0.15$) is constant.

The PMF is:

$$P(X = k) = (1 - p)^{k-1}p$$

Substitute $p = 0.15$ and $k = 4$:

$$P(X = 4) = (0.85)^3(0.15) = 0.6141 \cdot 0.15 = 0.0921$$



5 Hypergeometric Distribution

The hypergeometric probability formula is:

$$P(X = x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$$

Property	Formula / Value
PMF	$P(X = x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$
Support	$x = \max(0, n - N + K), \dots, \min(K, n)$
Mean	$n \frac{K}{N}$
Variance	$n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{N-n}{N-1}$
MGF	No closed-form
Example	Number of red balls drawn without replacement

Table 5: Characteristics of the Hypergeometric Distribution

- Population size: N , Successes in population: K
- Sample size: n
- Mean: $n \frac{K}{N}$
- Variance: $n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}$

Examples

- (i) $N = 20$, $K = 7$, $n = 5$, find $P(X = 2)$.

Solution: $P(X = 2) = \frac{\binom{7}{2} \binom{13}{3}}{\binom{20}{5}} = 0.3637$

- (ii) Find $E(X)$ and $Var(X)$ with $N = 30$, $K = 10$, $n = 6$.

Solution: $E(X) = 2$, $Var(X) \approx 0.97$

- (iii) A deck of 52 playing cards contains 12 face cards. If 5 cards are drawn randomly without replacement, what is the probability that exactly 2 of them are face cards?

Solution:

$$\begin{aligned}
 P(X = 2) &= \frac{\binom{12}{2} \binom{40}{3}}{\binom{52}{5}} \\
 &= \frac{66 \times 9880}{2598960} = 0.325
 \end{aligned}$$

- (iv) A company has 8 defective and 12 non-defective items. If 5 items are randomly selected without replacement, what is the probability that exactly 3 of them are defective?

Solution:

$$\begin{aligned} P(X = 3) &= \frac{\binom{8}{3} \binom{12}{2}}{\binom{20}{5}} \\ &= \frac{56 \times 66}{15504} = 0.238 \end{aligned}$$

A batch contains 10 defective and 40 non-defective items. If we randomly select 6 items, what is the probability that exactly 2 are defective?

Solution:

where: - $D = 10$ (defective items), - $N = 40$ (non-defective items), - $n = 6$ (sample size), - $k = 2$ (desired defective items).

$$\begin{aligned} P(X = 2) &= \frac{\binom{10}{2} \binom{40}{4}}{\binom{50}{6}} \\ &= \frac{\frac{10!}{2!(8!)} \times \frac{40!}{4!(36!)}}{\frac{50!}{6!(44!)}} \\ &= \frac{45 \times 91,390}{15,890,700} \\ &= 0.392 \end{aligned}$$

Thus, the probability is **0.392**.

- (v) A class has 25 students, 10 of whom are girls. If 6 students are selected at random without replacement, what is the probability that exactly 3 are girls?

Solution: This is a **Hypergeometric distribution** problem because:

- The population is finite ($N = 25$),
- There are $K = 10$ successes (girls) in the population,
- We are drawing $n = 6$ students without replacement.

The PMF is:

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

Substitute $N = 25$, $K = 10$, $n = 6$.

- (vi) A box contains 20 chocolates, of which 5 are dark chocolate. If 4 chocolates are selected at random without replacement, what is the probability that exactly 2 are dark chocolate?

Solution: This is a **Hypergeometric distribution** problem because:

- The population is finite ($N = 20$),
- There are $K = 5$ successes (dark chocolates) in the population,
- We are drawing $n = 4$ chocolates without replacement.

The PMF is:

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

Substitute $N = 20$, $K = 5$, $n = 4$, and $k = 2$:

$$P(X = 2) = \frac{\binom{5}{2} \binom{15}{2}}{\binom{20}{4}} = \frac{10 \cdot 105}{4845} = 0.2167$$

6 Multinomial Distribution

Property	Formula / Value
PMF	$P(X_1 = x_1, \dots, X_k = x_k) = \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}$
Support	$X_i \geq 0, \quad \sum X_i = n$
Mean	$E[X_i] = np_i$
Variance	$Var(X_i) = np_i(1 - p_i)$
Covariance	$Cov(X_i, X_j) = -np_i p_j$
MGF	$(\sum p_i e^{t_i})^n$
Example	Counts of votes among k candidates

Table 6: Characteristics of the Multinomial Distribution

- Generalization of binomial
- Trials: n , Categories: k
- Probabilities: $p_1, p_2, \dots, p_k$

Examples

- (i) $n = 5, p_1 = 0.5, p_2 = 0.3, p_3 = 0.2$, find $P(X_1 = 2, X_2 = 2, X_3 = 1)$.

Solution: $P = \frac{5!}{2!2!1!} (0.5)^2 (0.3)^2 (0.2) = 0.135$

- (ii) Find the probability when $n = 4, p_1 = 0.4, p_2 = 0.6, P(X_1 = 3, X_2 = 1)$.

Solution: $P = \frac{4!}{3!1!} (0.4)^3 (0.6) = 0.1536$

- (iii) A die is rolled 6 times. What is the probability of obtaining 2 ones, 1 two, 1 three, and 2 fours?

Solution:

Here, the possible outcomes are $\{1, 2, 3, 4, 5, 6\}$, and each face appears with probability $p_i = \frac{1}{6}$.

Given: $n = 6, x_1 = 2, x_2 = 1, x_3 = 1, x_4 = 2, x_5 = 0, x_6 = 0$.

Using the multinomial formula:

$$\begin{aligned}
 P(X_1 = 2, X_2 = 1, X_3 = 1, X_4 = 2, X_5 = 0, X_6 = 0) &= \frac{6!}{2!1!1!2!0!0!} \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^1 \left(\frac{1}{6}\right)^1 \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^0 \left(\frac{1}{6}\right)^0 \\
 &= \frac{720}{2!1!1!2!} \times \left(\frac{1}{6}\right)^6 \\
 &= \frac{720}{4} \times \frac{1}{46656} \\
 &= 180 \times \frac{1}{46656} \\
 &= 0.00386.
 \end{aligned}$$

- (iv) A box contains 3 different colors of balls: red, blue, and green. The probabilities of drawing each color are $p_1 = 0.5, p_2 = 0.3$, and $p_3 = 0.2$. If we draw 5 balls, what is the probability of drawing 2 red, 2 blue, and 1 green?

Solution:

Given: $n = 5, x_1 = 2, x_2 = 2, x_3 = 1$.

Using the multinomial formula:

$$\begin{aligned}
 P(X_1 = 2, X_2 = 2, X_3 = 1) &= \frac{5!}{2!2!1!} (0.5)^2 (0.3)^2 (0.2)^1 \\
 &= \frac{120}{2!2!1!} \times (0.25) \times (0.09) \times (0.2) \\
 &= \frac{120}{4} \times 0.0045 \\
 &= 30 \times 0.0045 \\
 &= 0.135.
 \end{aligned}$$

Thus, the probability is 0.135.

- (v) A survey categorizes customer feedback into three types: positive, neutral, and negative. The probabilities of each type are 0.6, 0.3, and 0.1, respectively. If 8 customers give feedback, what is the probability that 5 are positive, 2 are neutral, and 1 is negative?

Solution:

Given: $n = 8$, $x_1 = 5$, $x_2 = 2$, $x_3 = 1$.

$$\begin{aligned}
 P(X_1 = 5, X_2 = 2, X_3 = 1) &= \frac{8!}{5!2!1!} (0.6)^5 (0.3)^2 (0.1)^1 \\
 &= \frac{40320}{120 \times 2 \times 1} \times (0.07776) \times (0.09) \times (0.1) \\
 &= \frac{40320}{240} \times 0.00069984 \\
 &= 168 \times 0.00069984 \\
 &= 0.1176.
 \end{aligned}$$

Thus, the probability is 0.1176.

- (vi) A genetics experiment considers three possible traits: dominant (D), recessive (R), and neutral (N), with probabilities 0.4, 0.4, and 0.2, respectively. If 7 organisms are studied, what is the probability that 3 exhibit the dominant trait, 3 the recessive trait, and 1 the neutral trait?

Solution:

Given: $n = 7$, $x_1 = 3$, $x_2 = 3$, $x_3 = 1$.

$$\begin{aligned}
 P(X_1 = 3, X_2 = 3, X_3 = 1) &= \frac{7!}{3!3!1!} (0.4)^3 (0.4)^3 (0.2)^1 \\
 &= \frac{5040}{6 \times 6 \times 1} \times (0.064) \times (0.064) \times (0.2) \\
 &= \frac{5040}{36} \times 0.0008192 \\
 &= 140 \times 0.0008192 \\
 &= 0.1147.
 \end{aligned}$$

Thus, the probability is 0.1147.

- (vii) A multiple-choice test consists of 10 questions, each with 4 answer choices (A, B, C, D), and a student guesses all answers randomly. What is the probability that the student selects 4 A's, 3 B's, 2 C's, and 1 D?

Solution:

Given: $n = 10$, $x_1 = 4$, $x_2 = 3$, $x_3 = 2$, $x_4 = 1$, with each option having probability $p = 0.25$.

$$\begin{aligned}
P(X_1 = 4, X_2 = 3, X_3 = 2, X_4 = 1) &= \frac{10!}{4!3!2!1!} (0.25)^4 (0.25)^3 (0.25)^2 (0.25)^1 \\
&= \frac{3628800}{24 \times 6 \times 2 \times 1} \times (0.0009765625) \\
&= \frac{3628800}{288} \times 0.0009765625 \\
&= 12600 \times 0.0009765625 \\
&= 0.0123.
\end{aligned}$$

Thus, the probability is 0.0123.

- (viii) A game involves spinning a wheel with three equal sections labeled A, B, and C. A player spins the wheel 6 times. What is the probability that the outcomes are 3 A's, 2 B's, and 1 C?

Solution:

Given: $n = 6$, $x_1 = 3$, $x_2 = 2$, $x_3 = 1$, with each outcome having probability $p = \frac{1}{3}$.

$$\begin{aligned}
P(X_1 = 3, X_2 = 2, X_3 = 1) &= \frac{6!}{3!2!1!} \left(\frac{1}{3}\right)^3 \left(\frac{1}{3}\right)^2 \left(\frac{1}{3}\right)^1 \\
&= \frac{720}{6 \times 2 \times 1} \times (0.004115226) \\
&= \frac{720}{12} \times 0.004115226 \\
&= 60 \times 0.004115226 \\
&= 0.2469.
\end{aligned}$$

Thus, the probability is 0.2469.

7 Poisson Distribution

Property	Formula / Value
PMF	$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$
Support	$X \in \{0, 1, 2, \dots\}$
Mean	λ
Variance	λ
MGF	$e^{\lambda(e^t - 1)}$
Cumulants	$\kappa_n = \lambda$ for all n
Example	Number of arrivals per time unit

Table 7: Characteristics of the Poisson Distribution

- Number of events in fixed interval
- Parameter: λ
- Mean: λ
- Variance: λ

Examples

- (i) For $\lambda = 5$, find $P(X = 3)$.

Solution: $P(X = 3) = \frac{5^3 e^{-5}}{3!} \approx 0.1404$

- (ii) For $\lambda = 4$, find $P(X = 0)$.

Solution: $P(X = 0) = e^{-4} \approx 0.0183$

- (iii) The average number of emails received per hour is 5. What is the probability of receiving exactly 3 emails in an hour?

Solution:

Given $\lambda = 5$, $x = 3$:

$$\begin{aligned} P(X = 3) &= \frac{e^{-5}5^3}{3!} \\ &= \frac{e^{-5} \times 125}{6} \\ &= 0.1404. \end{aligned}$$

- (iv) A factory records an average of 2 machine failures per day. What is the probability that there are no failures in a day?

Solution:

Given $\lambda = 2$, $x = 0$:

$$\begin{aligned} P(X = 0) &= \frac{e^{-2}2^0}{0!} \\ &= e^{-2} \\ &= 0.1353. \end{aligned}$$

- (v) A website receives an average of 10 visitors per minute. What is the probability of receiving exactly 7 visitors in a given minute?

Solution:

Given $\lambda = 10$, $x = 7$:

$$\begin{aligned} P(X = 7) &= \frac{e^{-10}10^7}{7!} \\ &= 0.0902. \end{aligned}$$

- (vi) A call center receives an average of 3 calls per hour. What is the probability that exactly 5 calls occur in an hour?

Solution:

Given $\lambda = 3$, $x = 5$:

$$\begin{aligned} P(X = 5) &= \frac{e^{-3}3^5}{5!} \\ &= 0.1008. \end{aligned}$$

- (vii) A train station experiences an average of 4 train arrivals per hour. What is the probability of exactly 6 arrivals in an hour?

Solution:

Given $\lambda = 4$, $x = 6$:

$$\begin{aligned} P(X = 6) &= \frac{e^{-4}4^6}{6!} \\ &= 0.1042. \end{aligned}$$

- (viii) The number of misprints in a newspaper follows a Poisson distribution with an average of 1 misprint per page. What is the probability that a randomly selected page has exactly 2 misprints?

Solution:

Given $\lambda = 1$, $x = 2$:

$$\begin{aligned} P(X = 2) &= \frac{e^{-1}1^2}{2!} \\ &= 0.1839. \end{aligned}$$

- (ix) A rare disease occurs in a population at an average rate of 0.5 cases per year. What is the probability that there will be no cases in a year?

Solution:

Given $\lambda = 0.5$, $x = 0$:

$$\begin{aligned} P(X = 0) &= e^{-0.5} \\ &= 0.6065. \end{aligned}$$

- (x) A traffic light at an intersection experiences an average of 8 cars passing per minute. What is the probability that exactly 10 cars pass in a given minute?

Solution:

Given $\lambda = 8$, $x = 10$:

$$\begin{aligned} P(X = 10) &= \frac{e^{-8} 8^{10}}{10!} \\ &= 0.0994. \end{aligned}$$

- (xi) A bookstore sells an average of 15 books per day. What is the probability that exactly 20 books are sold in a day?

Solution:

Given $\lambda = 15$, $x = 20$:

$$\begin{aligned} P(X = 20) &= \frac{e^{-15} 15^{20}}{20!} \\ &= 0.0516. \end{aligned}$$

- (xii) A customer service center receives an average of 12 complaints per day. What is the probability of receiving exactly 8 complaints in a day?

Solution:

Given $\lambda = 12$, $x = 8$:

$$\begin{aligned} P(X = 8) &= \frac{e^{-12} 12^8}{8!} \\ &= 0.0653. \end{aligned}$$